Implementing a new Galaxy integration to export datasets to COPO

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Background

Galaxy tools and **workflows** [doi:10.1038/s41587-023-02100-3] have recently been developed to support international **biodiversity** efforts like:

- the Vertebrate Genomes Project (VGP) https://vertebrategenomesproject.org/
- the European Reference Genome Atlas (ERGA) https://www.erga-biodiversity.eu/
- the Darwin Tree of Life (ToL) https://www.darwintreeoflife.org/

The common goal of these initiatives is to generate high-quality reference **genome assemblies** in large scale.

These Galaxy workflows are able to produce genome assemblies and annotations starting from the sequencing reads, but often a **manual step** is required at the end of this process to submit these large data (together with the required metadata) to a deposition database like the European Nucleotide Archive (ENA).

This last step can be facilitated by using the COPO brokering portal, which guides the user in the submission process while ensuring that all the necessary metadata are provided.

FAIR principles

The FAIR principles [doi:10.1038/sdata.2016.18], published in 2016, provide community-developed guidelines to improve the Findability, Accessibility, Interoperability, and Reusability of **digital objects** by both humans and machines.

These principles rely on recording high-quality **metadata**, which must meet the requirements and understanding of many stakeholders in order to be truly useful.

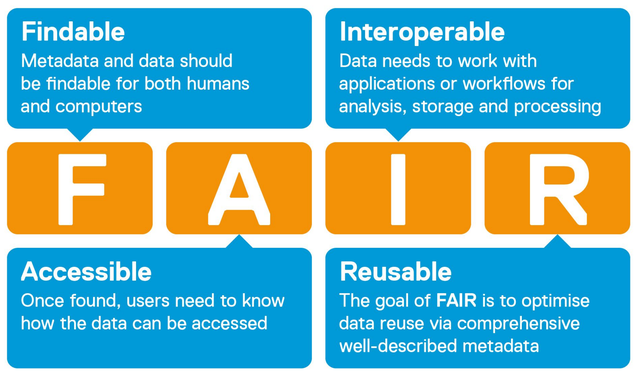


Image source: https://scibite.com/solutions/enterprise-fair-data-mdm/

Metadata and Data on local computer

H	Lt	loc
1.3	16	NWI
1.34	16	BEL
...	...	...

Metadata and Data on institutional web server

Uid	Height	Hours of light per day	Location
Nfj123	1.3	16	Norwich
Nfj523	1.34	16	Belfast
...	...	...	

Validated community derived metadata and data, deposited to a public repository

Unique ID: NCIT_C164823	Height: NCIT_C25347	Daylight: EOL_0001659	Geographic location: GAZ_00000448
8890	1.3	16	Norwich: GAZ_00004867
8891	1.34	16	Belfast: GAZ_22222459
...	...	...	...

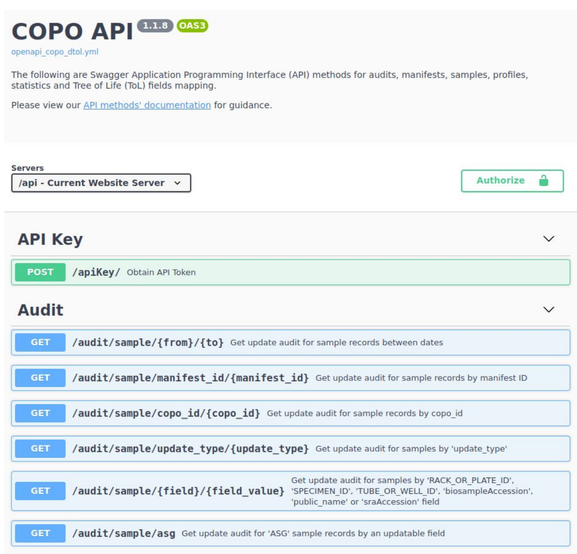
Increasing FAIRness

Galaxy-COPO integration

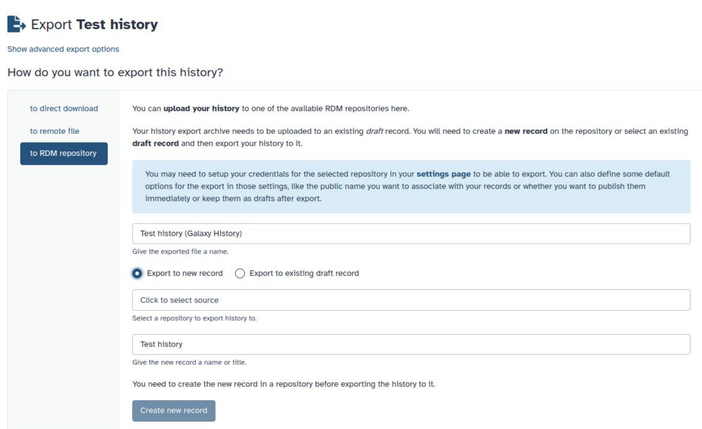
To integrate the currently separate processes of production and deposition of genome assemblies, we are working on extending:

- the Galaxy backend code and user interface
- the COPO Application Programming Interface (API)

to allow users to easily **export** their Galaxy datasets to the COPO platform. From there, they can then start the metadata validation and submission process to ENA.



https://copo-project.org/api/



https://galaxyproject.org/news/2024-05-03-inveniordm-integration/

Users will be able to add their COPO API token into your Galaxy user preferences, under "Manage Information".

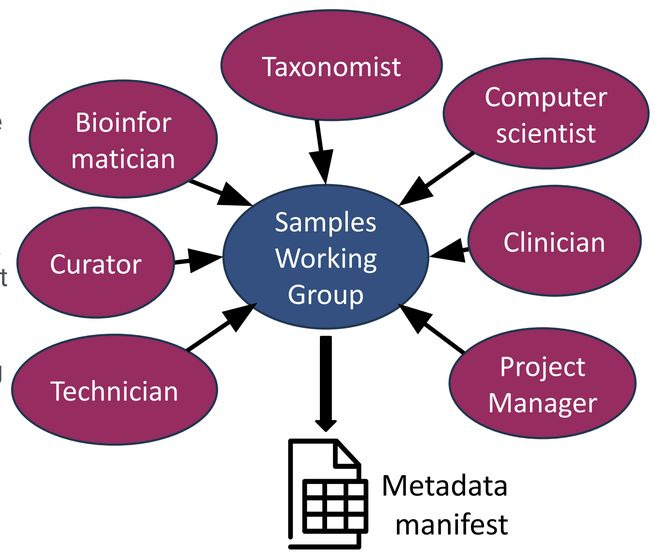
Then the dataset export mechanism will be similar to the already existing Galaxy integration with Invenio RDM, developed by David López.

Once completed, this integration between Galaxy and COPO will improve the interoperability of both projects, and simplify FAIR data management for users.

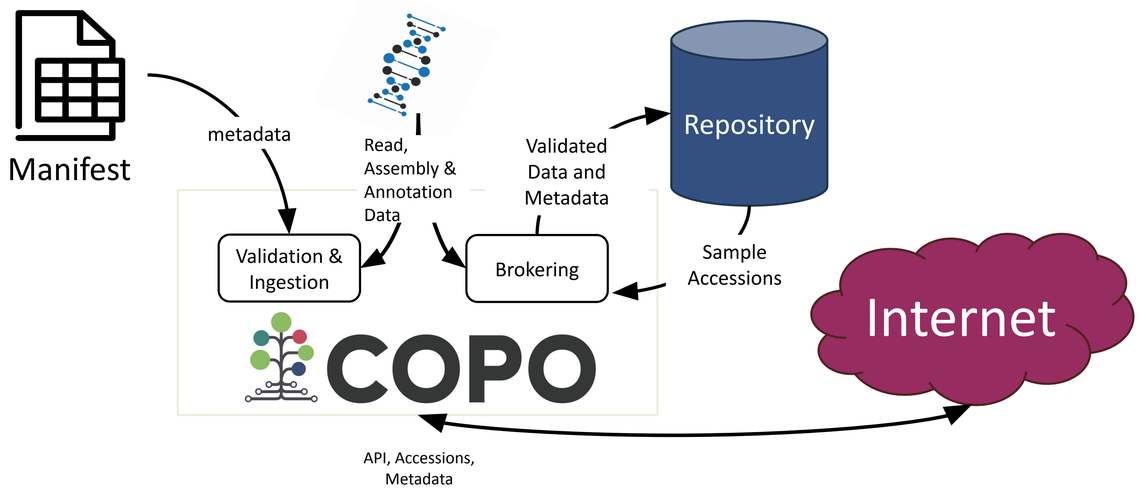
COPO

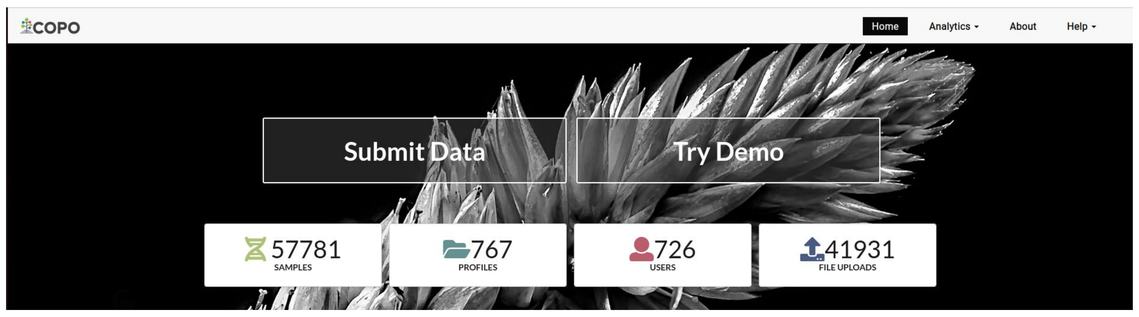
With COPO (Collaborative OPen Omics) and the Darwin ToL and ERGA projects, we have sought as broad a consensus as possible in designing **metadata standards** for biodiversity [doi:10.12688/wellcomeopenres.18499.3]. This provides a plurality of opinion on what constitute important metadata.

This can be a difficult and time-consuming process, due to scheduling, communication, ideology, and politics. Nevertheless, it essential to get this right.



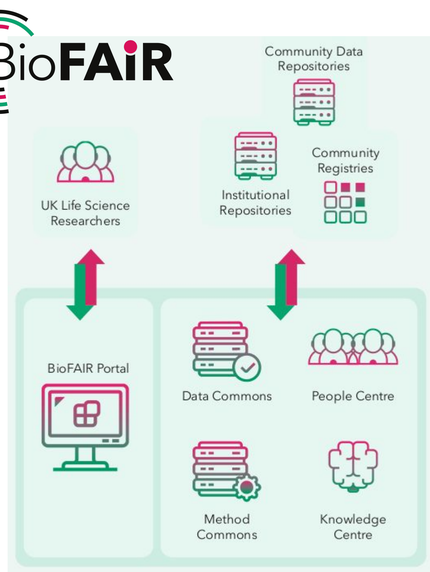
COPO [doi:10.12688/f1000research.23889.1] is a web service which accepts, **validates** and **brokers** standardised metadata, along with associated data (e.g. sequence reads, genome assemblies, image etc).
COPO brokers these data into a variety of repositories, such as the European Nucleotide Archive (ENA), Biosamples and the Bioimage Archive. This ensures quality of metadata and FAIRness, whilst reducing the burden on researchers.





BioFAIR

This project is sponsored by BioFAIR (https://biofair.uk/), a new federated **digital research infrastructure** conceived by ELIXIR-UK to address the urgent need for more effective Research Data Management (RDM) and reproducible data processing **workflows**.
BioFAIR's goal is to provide end-to-end **FAIR data management** and analysis capabilities along with the necessary support and training for UK researchers working in the life sciences.
In 2023, UK Research and Innovation (UKRI) announced a five-year funding of £34 million for BioFAIR, subject to business case approvals.
A Roadshow across the UK is currently underway to engage the community and shape the initial priorities for BioFAIR.



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Cellular Genomics

Decoding Biodiversity